29. Morphological Operations Using OpenCV – Erosion Technique

Aim

To perform **morphological erosion** on an image using OpenCV and observe the effect of erosion on the image.

Procedure

1. Read the input image using OpenCV.
2. Convert the image into grayscale.
3. Create a structuring element (kernel).
4. Apply the **erosion** operation using cv2.erode().
5. Display the original and eroded images.
6. Save the resulting image.

Program

```
import cv2

import numpy as np


# Read input image

img = cv2.imread("input.jpg")


# Check image

if img is None:

    print("Error: Image not found")

    exit()


# Convert to grayscale
```

```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)


# Create structuring element
kernel = np.ones((5, 5), np.uint8)


# Apply erosion
eroded = cv2.erode(gray, kernel, iterations=1)


# Display images
cv2.imshow("Original Image", gray)
cv2.imshow("Eroded Image", eroded)


# Save output
cv2.imwrite("eroded_image.jpg", eroded)


cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input

Output